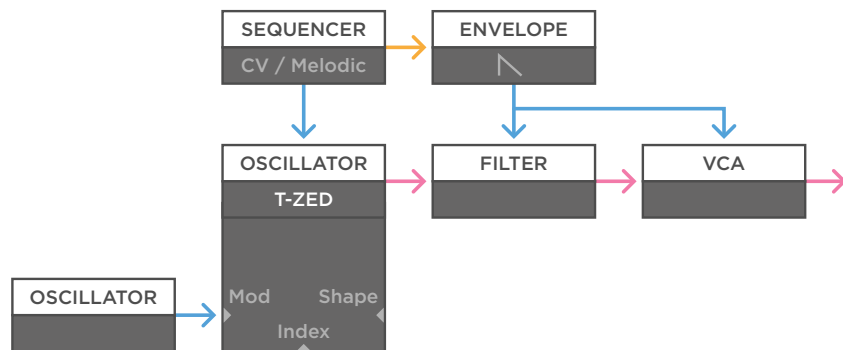


Phase Modulation

LA67 T-ZED

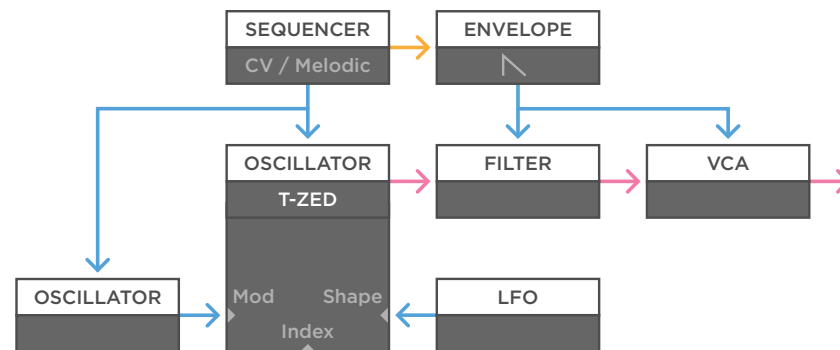
- Audio
- Control Voltage
- Trigger / Gate



(a)

a - In this patch, phase modulation is used in a simple voice, by sending the T-ZED oscillator through a filter and VCA. Use a sequencer to tune the T-ZED, and trigger a simple envelope to open the filter and VCA. Subtle phase modulation amounts will result in subtle textures. While strong modulation with harsher waveforms can be interesting for more metallic, experimental and percussive sounds.

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(b)

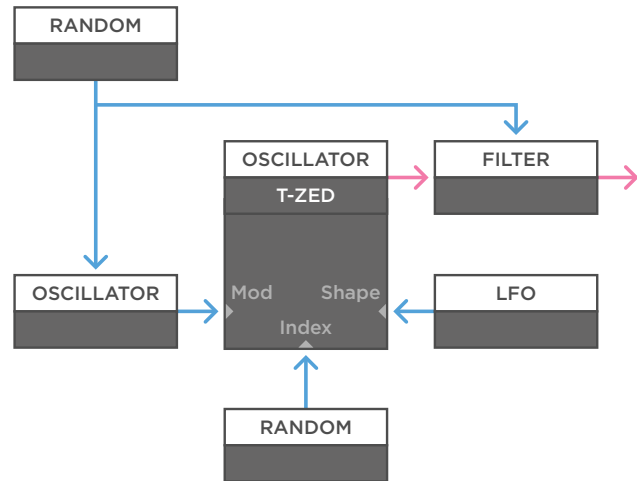
b - A lot of tonal textures can be achieved when the oscillator used to modulate the Phase, is tuned and sequenced as well. You can multiply the 1v/oct signal going to the T-ZED, and use it to sequence the modulating oscillator.

With the T-ZED offering wave-shape modulation as well, you can easily increase the sonic variations over time. For example, by sending a slow LFO or random voltage to the shape input.

Phase Modulation

LA67 T-ZED

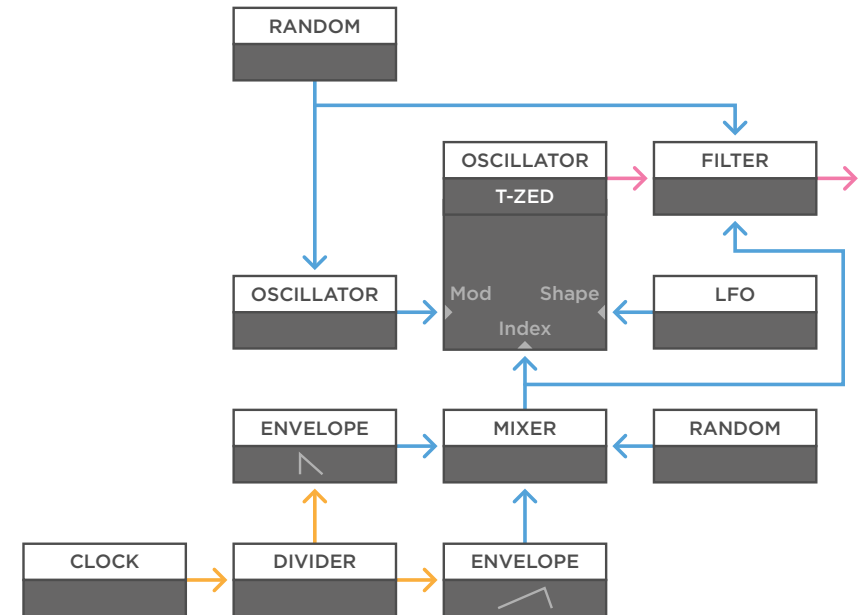
- Audio
- Control Voltage
- Trigger / Gate



(a)

a - You can create interesting drones and textures by sending a random voltage to the Index. You can expand the depth by sending another random voltage to the frequency of the modulating oscillator, and multiply that signal to the filter. To create slowly changing sounds over time, add a slow random voltage to the index, and an LFO to the shape input.

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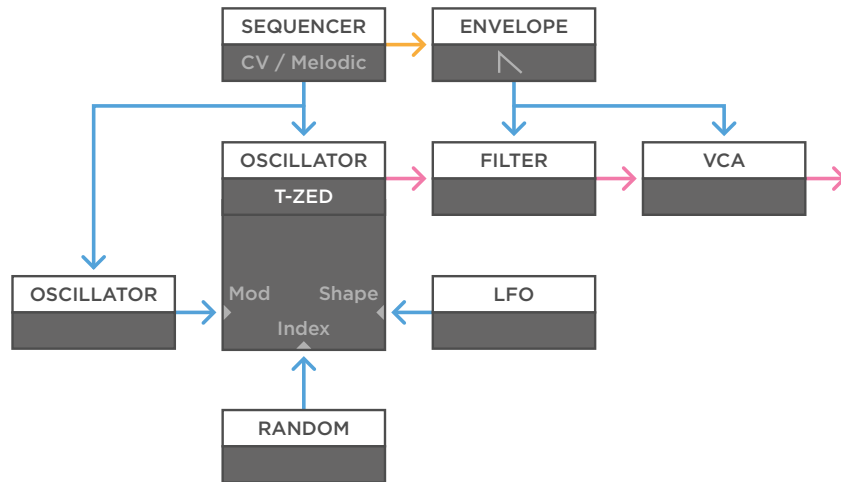
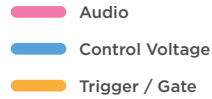


(b)

b - You can alter this patch and add interesting tempo synced modulation by replacing the random voltage to the Index for a mixer, and use a clock and clock divider to trigger two different envelopes. Mix them together with the random voltage, and use them on the Index, as well as the filter.

Phase Modulation

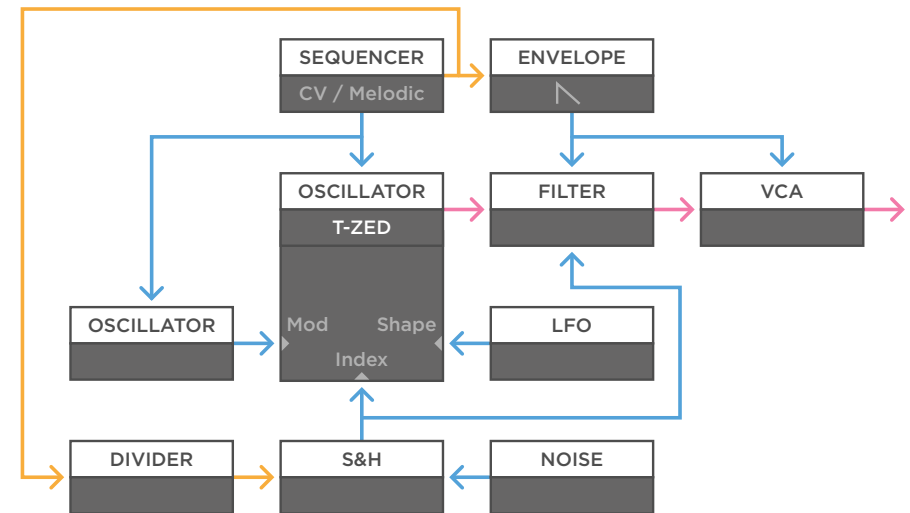
LA67 T-ZED



(a)

a - Modulating the Index for time based variation can be applied to the simple voice shown on the first page as well. Just add a slow LFO or random voltage to the Index, to create subtle changes in texture over time.

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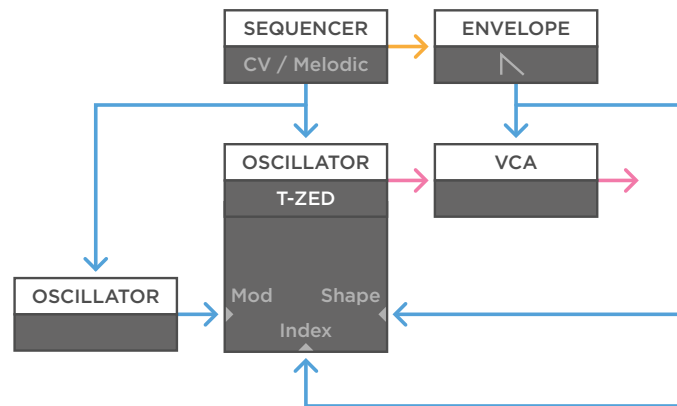
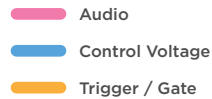


(b)

b - You can create interesting tempo synced variations to this by multiplying the clock from the sequencer to a clock divider, and then use a division to trigger an envelope, or a sample and hold signal for example. Again, you can multiply that signal and send it to the filter for some added depth.

Phase Modulation

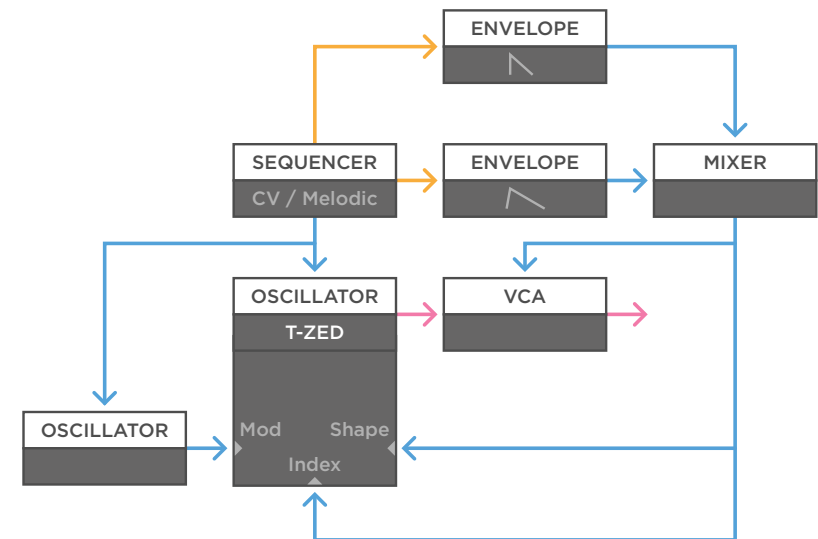
LA67 T-ZED



(a)

a - The index cv gives you the option to add a lot of harmonics to your waveforms without other wave shapers or filters. In this simple setup a sequencer is used to tune both the T-ZED, and the modulating oscillator. It also triggers a simple envelope to open the VCA. You can multiply that envelope, and send it to the Index, to control the phase modulation amount. And if you like, even send a copy to the shape input as well, to create harsher sounds.

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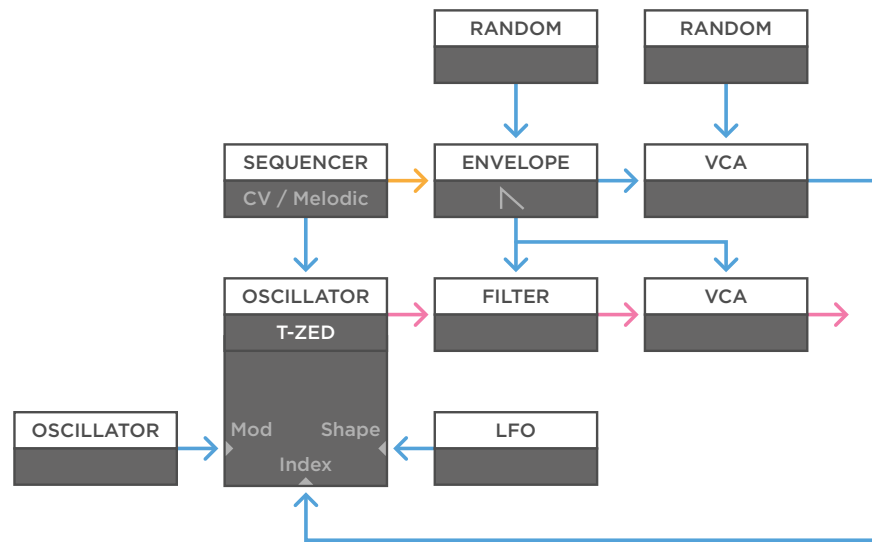
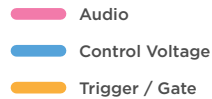


(b)

b - You can expand the setup further by combining more than one modulation source. For example, beside using the gate from the sequence, use the steady clock from the sequencer to trigger a second envelope with different settings. Then, use a mixer to mix different amounts of those envelopes together, before going to the VCA, shape and index. Tweaking the mixer by hand will give some interesting performance options.

Phase Modulation

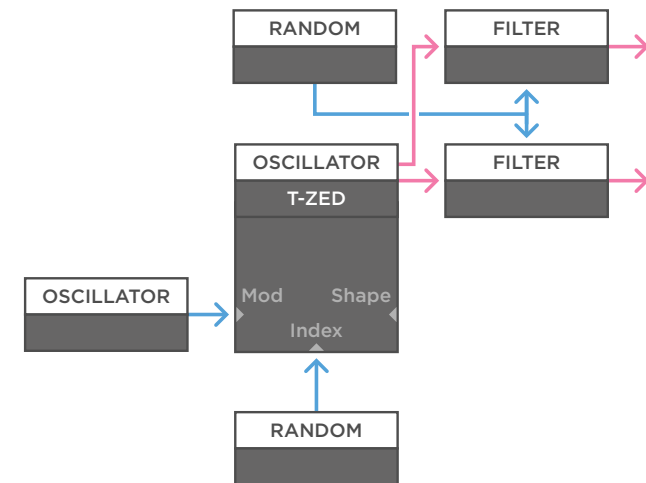
LA67 T-ZED



(a)

a - Despite the built-in VCA, it can be worth it to experiment with more modulation, before modulating the Index. In this basic setup for example, you can send a copy of the envelope to an external VCA, before going on to the Index. Then you can modulate the amount of envelope passing on, by sending something like a random voltage to that VCA. To add some extra dynamics, modulate the decay of the envelope with another random voltage.

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(b)

b - Having multiple waveform outputs, gives you the possibility to easily make stereo patches. This sounds great with phase modulation. For example take the square output to one filter, and the saw output to another. Use an oscillator for Phase mod, a random voltage to the modulation amount, and something like another random voltage to modulate both filters, for some extra movement.